Project: Molecular dynamics simulation of a one-dimensional Lennar-Jones system

Objective: To study the structure and dynamics of one-dimensional Lennard-Jones systems.

Outline

1. Write a MD code for a simulation of a Lennard-Jones (LJ) system in one dimension. Use N = 100 atoms. Remember that the LJ potential is of the form: $U_{LJ}(r) = 4\varepsilon[(\sigma/r)^{12} - (\sigma/r)^6]$.
2. Create an initial set-up of positions from the sites of a one-dimensional lattice.
3. Create an initial set-up of velocities from a Maxwell distribution. Remember that the velocity of the center of mass of the system must be zero.
4. Run a MD simulation for argon at T^* =1.0 and ρ^*= 0.95. Re-scale the temperature if necessary. Compute the interaction energy U^*, the pressure P^*, the radial distribution function $g(r)$, the velocity autocorrelation function $C(t)$ and the mean square displacement $r^2(t)$ of the system.
5. Repeat the same study for a system of soft spheres, with an interaction potential given by

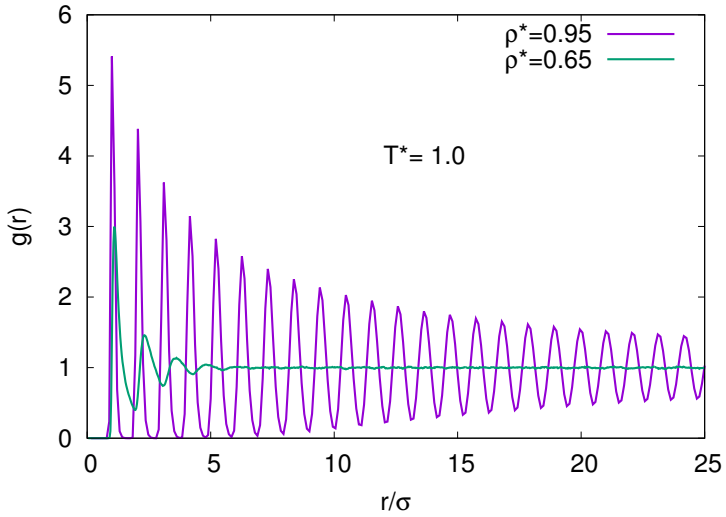
$$U_{ss}(r) = \begin{cases} U_{LJ}(r) + \varepsilon, & 0 < r \leq 2^{1/6}\sigma \\ 0, & r > 2^{1/6}\sigma \end{cases}$$

6. Compare the results you have obtained by using the different interaction potentials.
7. Repeat the simulations for the following state points: $T^* = 1.0$; $\rho^* = 0.65$ and $T^* = 1.0$; $\rho^* = 0.35$.

Lennard-Jones parameters

atom	mass(uma)	ϵ/k_B (K)	σ ($\mathring{A}$)
Ar	39.948	119.8	3.405

Some results to compare with:



Radial distribution functions for the Lennard-Jones system at different densities.

Questions

1. Give the expressions of the pressure P and the radial distribution $g(r)$ for the case of a one-dimensional system.

2. Define the reduced units used in your simulations. Translate from reduced units to real units the values of T^*, ρ^*, U^* and P^*.

3. Discuss the definition given for the soft sphere interaction potential.

4. Give the expression of the self-diffusion coefficient D fot the case of a one-dimensional system. Discuss the results obtained for different densities and different interaction potentials.